Chapter 12
SDG 12: Responsible Consumption and Production: Using PBL to Foster Engagement and Empower Students with Sustainable Practices

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Abstract Project-Based Learning (PBL) is an educational approach that emphasizes learning through active engagement in real-world projects. It involves students working collaboratively on projects that are relevant and meaningful, allowing them to develop critical thinking, problem-solving, and communication skills. PBL promotes inquiry, research, and application of knowledge in a practical context, fostering a deeper understanding of the subject matter. Supporting developing countries to strengthen their scientific and technological capacity is crucial for their sustainable development. By enhancing their capabilities in science and technology, these countries can address challenges related to sustainable patterns of consumption, production, and transportation. This support can involve various aspects such as knowledge transfer, technology transfer, capacity building programmes, research collaborations, and financial assistance. The authors believe that one of the most efficient ways to achieve societal change and sustainable actions that may perdure is through education. Therefore, by empowering students in scientific and technological fields, they can adopt more sustainable practices, innovate solutions, and contribute to global efforts in achieving sustainable development goals wherever they will work and be. The addressed problem presented in this chapter deals with solid waste transportation logistics, optimizing routes, and ensuring the use of environmentally

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friendly modes of solid waste transportation management through the utilization of multi-criteria decisions. Addressing these problems requires a holistic approach that includes investment in sustainable infrastructure, collaboration with transportation providers, adoption of new technologies, policy support, and employee engagement initiatives. By overcoming these challenges, companies can contribute to reducing the environmental impact of transportation and move towards more sustainable practices. At the same time, students involved in their solution proposals acquire competencies that they might apply in their future working careers. These projects are developed by young master students at the Polytechnic University of Coimbra, Coimbra Engineering Institute and the methodology used was a study case approach. These students belong to Gen Z, meaning those born since 1995. This generation is characterised by hyper-connected, agile, individualistic, simple, tolerant individuals who embrace diversity. Named after alpha, the first letter in the Greek alphabet, Generation Alpha is the first to be born entirely in the twenty-first century and the third millennium. Generation Alpha will confront unique circumstances, particularly environmental challenges such as extreme weather events, which are projected to increase in prevalence during their lifetimes. This problem-based project proposal was immediately accepted by them and at its conclusion. In conclusion, we believe that the students have gained experience and a desire to continue to study these subjects and are more aware of how their knowledge can make a difference in people's daily lives and practices. Stakeholders will be informed, and new improvements can be made collaboratively.

Keywords Sustainable Development Goal 12 · Problem-based learning · Sustainable consumption · Sustainable transportation students engagement

1 Introduction and Problem Definition

This chapter is directly aligned with Sustainable Development Goal 12 (SDG 12)—Responsible Consumption and Production, which aims to promote sustainable management and efficient use of natural resources, reduce waste generation, and encourage environmentally responsible practices. By focusing on optimizing solid waste transportation logistics, the project contributes to Target 12.5, which seeks to substantially reduce waste generation through prevention, reduction, recycling, and reuse. The use of multi-criteria decision-making models supports more efficient resource allocation, ensuring that waste collection processes are both cost-effective and environmentally sustainable. Furthermore, the integration of Problem-Based Learning (PBL) into this initiative empowers students with the skills and knowledge needed to develop innovative solutions, aligning with Target 12.8, which emphasizes the importance of ensuring that people have the relevant information and awareness for sustainable development and lifestyles in harmony with nature. By engaging students in real-world problem-solving and collaborating with stakeholders, this

project reinforces the role of education in fostering sustainability, demonstrating how academic initiatives can drive practical environmental and social change.

To perform efficient waste management, it is necessary to have a good collection system, therefore, there is a need to collect a good amount of waste in a short period to reduce transport costs and increase collection capacity. The study area chosen to carry out this service ensuring a better life quality for its residents starts at União de Freguesias de Eiras and ends at ERSUC—Resíduos Sólidos do Centro, S.A, which is a Centre of Solid Residue Treatment.

Collection systems must have a multidimensional and collaborative approach, making them closer to residents, considering the urban conditions of the territories, operation and maintenance costs, and capacity. The authors highlight some important points such as:

- Prior planning: Before starting a collection, carry out detailed planning of the area that will be covered, identifying the critical points where the waste is concentrated.
- Use of technology: Using technology, such as mapping applications or monitoring systems, to identify critical areas and optimize collection routes.
- Suitable vehicles: using transport vehicles with a good load capacity to collect waste more efficiently.
- Continuous assessment: Regular assessments of collection progress, ensuring that all points selected for collection are addressed.

Aiming at all the important points mentioned above, the chosen area begins at the Junta de Freguesia de Eiras in Coimbra-PT, which has several linked streets marked as a collection route area, dividing this area into several routes where the truck will pass to collect waste, taking into account the types of residences, commercial stores and even supermarkets along the route. An average of containers needed for the chosen area was thus made, considering those that already exist and those that will need to be included on the route to the final destination ERSUC (Empresa de Resíduos Sólidos Urbanos do Centro—Urban Solid Waste Company of the Centre of Portugal). Thus, establishing the resolution of two problems that are: the shortest route and maximum capacity route, determining the best container for the area according to the evaluation criteria.

1.1 Evaluation Criteria

To evaluate the best decision regarding waste collection, the following four criteria were chosen:

1. Final container prices: Final container prices are a relevant criterion that has direct impacts on the operational costs and economic viability of these systems. When container prices are high, the initial investment costs to implement or expand waste management systems can increase. Containers purchased must be

durable, high quality, and suitable for the specific waste collection needs of the community.

2. Containers' capacity: Another criterion is the capacity of recycling bins, that is, the places that deal with the amount of waste generated by residents. The importance is due to the fact that there are few recycling bins, or they are inadequate, and, as a result, the capacity can be overloaded, leading to problems such as the accumulation of uncollected waste, environmental pollution, and risks to public health.
3. Maintenance costs: Maintenance costs are a part of the operational costs associated with waste management systems and the maintenance of recycling bins. These costs include cleaning expenses, and repairing and replacing infrastructure such as recycling containers, compaction equipment, and collection vehicles. Lack of adequate maintenance can lead to deterioration of existing infrastructure, resulting in long-term additional costs. Thus, maintenance costs must be considered as part of the evaluation criteria to ensure the sustainability and effectiveness of waste management systems.
4. Collection and final discharge transportation time: Collection and final discharge transportation time is an important criterion to be considered when evaluating the efficiency of waste management systems. This time refers to the period necessary to collect waste from recycling bins and transport it to be processed or discharged into proper facilities. When the recycling bins are in spaced areas, transportation time can be significantly longer due to the need to travel longer distances to collect waste. This can result in higher operating costs, increased emissions of vehicle pollutants, and reduced efficiency in waste management, thus contributing to the deterioration of the quality of the environment, especially if there is a risk of leakage or spillage during transport.

2 Criteria Selection Justification

2.1 Containers Capacity

As shown in Table 12.1, the total area distributed over eight routes has a capacity of:

- 3300 kg distributed in size P (small) containers;
- 18,000 kg distributed in size M (medium) containers;
- 264,000 kg distributed in size G (large) containers.

2.2 Container's Price

As shown in Table 12.2, the total area distributed over the eight routes has a total cost of:

Table 12.1 Containers' capacity

Size	Capacity (kg)	Actual quantity	New quantity	Total
P	75	29	15	44
M	500	25	11	36
G	800	16	17	33

Table 12.2 Containers' price

Containers' size	Price per unit	Installed quantity	Total cost
P	160€	15	2400€
M	380€	11	4180€
G	800€	17	13,600€

Table 12.3 Maintenance costs

Size	Total quantity	Total cost
P	44	100€
M	36	250€
G	33	380€

- 2400 € for size P containers;
- 4180 € for size M containers;
- 13,600 € for size G containers.

2.3 Maintenance Costs

As shown in Table 12.3, the total area distributed over eight routes has maintenance costs:

- 100 € for a container set of size P;
- 250 € for a container set of size M;
- 380 € for a container set of size G.

2.4 Collection and Final Discharge Transportation Time

The collection time from the initial collection point to the final destination was defined in eight routes, which are divided as follows:

Route 1: corresponds to the shortest route, with 8.8 km. This path starts in União das Freguesias de Eiras and São Paulo de Frades, at Rua Dr. Alfredo de Freitas

(14 houses = 1P), following along Rua 25 de Abril (22 houses =2P), Rua Viseu (22 houses = 2P), Rua Rego (10 houses = 1P), Travessa Rego (2 houses = 1P), entering again in Rua Dr. Alfredo de Freitas (7 houses = 1P), Rua Cruz Nova (7 houses, 2 commercial points, 5 apartment blocks and 4 warehouses = 2 M + 1G), Rua Chaves (2 apartment blocks, 7 houses, 1 car dealer and 1 warehouse = 2 M + 1G), = Total 14 (8P + 4 M + 2G) (Path 1–2) passing through the roundabout, entering in Rua Barca Ribeira de Eiras (2 supermarket, 8 houses, 4 apartment blocks, 2 warehouses, 1 shop = 1 M + 2G), = Total 3 (Path 2–4) passing through other roundabout, entering at Estrada de Eiras (14 apartment blocks, 2 shops, 10 warehouses, 2 car dealers, 1 supermarket, 4 houses = 3P + 2 M + 6G), = Total 11 (Path 4–7) passing through another roundabout, passing through N111 and EN111 (20 houses, 3 shops = 2P+ 2 M+ 2G) = Total 6 (Path 7–9), a roundabout. At Rua da Liberdade (22 houses, 3 shops = 2P + 2 M + 1G) = Total 5 (Path 9–10), in Rua Quinta da Zombaria = 0 (Path 10–12), roundabout at Rua Coimbra until arriving ERSUC = 0 (Path 14–16).

Route 2: corresponds to a path with 12.3 km. The route begins in União das Freguesias de Eiras and São Paulo de Frades, at Rua Dr. Alfredo de Freitas (14 houses = 1P), following along Rua 25 de Abril (22 houses = 2P), Rua Viseu (22 houses =2P), Rua Santo Cristo (24 houses = 2P), Rua Montargil (28 houses = 1P + 1 M), Rua Cruz do Vale do Seixo (25 houses, 1 apartment block = 2 M + 2G) = Total 13 (Path 1–3), passing through the roundabout, Rua Vale de São Miguel (7 houses and 22 apartment blocks, 2 warehouses = 2P + 1 M + 1G), Bairro São Miguel (19 apartment blocks = 2P + 2 M + 1G), passing through the roundabout = Total 9 (Path 3–6), Estrada de Eiras (5 apartment blocks, 1 shopping centre, 5 car dealers, 7 warehouses = 3P + 3 M + 1G) = Total 7 (Path 6–5), passing through other roundabout, Rua Bento de Jesus Caraça, N1, IC2, IP3, E801, Rua da Quinta da Zombaria, passing the roundabout and arriving to ERSUC = Total 0 (Path 14–16).

Route 3: corresponds to a path with 10.7 km. The route begins in União das Freguesias de Eiras and São Paulo de Frades, at Rua Dr. Alfredo de Freitas (14 houses = 1P), following along Rua 25 de Abril (22 houses = 2P), Rua Viseu (22 houses = 2P), Rua Santo Cristo (24 houses = 2P), Rua Montargil (28 houses = 1P + 1 M), Rua Cruz do Vale do Seixo (25 houses, 1 apartment blocks = 2 M + 2G) = Total 13 (Path 1–3), passing through the roundabout, Rua Vale de São Miguel (7 houses and 22 apartment blocks, 2 warehouses = 2P + 1 M + 1G), Bairro São Miguel (19 apartment blocks = 2P + 2 M + 1G) = Total 9 (Path 3–6), passing through the roundabout, Estrada de Eiras (5 apartment blocks, 1 shopping centre, 5 car dealer, 7 warehouses = 3P + 3 M + 1G) = Total 7 (Path 6–5), passing through other roundabout, Rua Bento de Jesus Caraça, N1, IC2, roundabout, N336 (1 shop =1 M), other roundabout, Rua Coimbra, roundabout, Rua da Junqueira de Trouxemil (14 houses = 1P), Rua dos Jacintos (3 houses = 1P), Rua do Barreiro (19 houses, 1 warehouse = 1 M), Rua Almas (6 houses = 1P) = 3P + 2 M = Total 5 (Path 11–13) Largo Alma, Rua do Bolregão (1 house = 1P) = Total 1 (Path 13–14) arriving to ERSUC = Total 0 (Path 14–16).

Route 4: corresponds to another shortest path with 8.8 km. The route begins in União das Freguesias de Eiras and São Paulo de Frades, at Rua Dr. Alfredo de Freitas (14 houses = 1P), following through Rua 25 de Abril (22 houses =2P), Rua Viseu (22 houses = 2P), Rua Rego (10 houses = 1P), Travessa Rego (2 houses = 1P), entering again in Rua Dr. Alfredo de Freitas (7 houses = 1P), Rua Cruz Nova (7 houses, 2 shops, 5 apartment blocks and 4 warehouses = 2 M + 1G), Rua Chaves (2 apartment blocks, 7 houses, 1 car dealer and 1 warehouse = 2 M + 1G) = Total 14 (8P + 4 M + 2G) (Path 1–2), passing through the roundabout, entering at Rua Entre Vinhas (4 warehouses, 1 supermarket, 1 car dealer, 4 houses = 2P + 2 M + 2G), Estrada de Eiras (4 warehouses, 1 car dealer, 22 commercial buildings, 2 apartment blocks, 8 houses = 3P + 2 M +4G) = Total 15 (Path 2–5), following E801 = Total 0 (Path 5–14), arriving at ERSUC = Total 0 (Path 14–16).

Route 5: corresponds to another shortest path with 8.8 km. The route begins in União das Freguesias de Eiras and São Paulo de Frades, at Rua Dr. Alfredo de Freitas (14 houses = 1P), following through Rua 25 de Abril (22 houses =2P), Rua Viseu (22 houses = 2P), Rua Rego (10 houses = 1P), Travessa Rego (2 houses = 1P), entering again in Rua Dr. Alfredo de Freitas (7 houses = 1P), Rua Cruz Nova (7 houses, 2 shops, 5 apartment blocks and 4 warehouses = 2 M + 1G), Rua Chaves (2 apartment blocks, 7 houses, 1 car dealer and 1 warehouse = 2 M + 1G), = Total 14 (8P + 4 M + 2G) (Path 1–2), passing through the roundabout, entering in Rua Entre Vinhas (4 warehouses, 1 supermarket, 1 car dealer, 4 houses = 2P + 2 M + 2G) following to Estrada de Eira (4 warehouses, 1 car dealer, 22 commercial buildings, 2 apartment blocks, 8 houses = 3P + 2 M +4G) = Total 15 (Path 2–5), N1, IC2 = Total 0 (Path 5–8) passing through the roundabout, N336 (1 shop =1 M) = Total 1 (Path 8–11), roundabout, Rua Coimbra, roundabout, Rua da Junqueira de Trouxemil (14 houses = 1P), Rua dos Jacintos (3 houses = 1P), Rua do Barreiro (19 houses, 1 warehouse = 2 M), Rua Almas (6 houses = 1P) = Total 5 (3P + 2 M) (Path 11–13) Largo Alma, Rua do Bolregão (1 house = 1P) = Total 1 (Path 13–14) arriving to ERSUC = Total 0 (Path 14–16).

Route 6: corresponds to another shortest path with 8.8 km. The route begins in União das Freguesias de Eiras and São Paulo de Frades, at Rua Dr. Alfredo de Freitas (14 houses = 1P), following through Rua 25 de Abril (22 houses =2P), Rua Viseu (22 houses = 2P), Rua Rego (10 houses = 1P), Travessa Rego (2 houses = 1P), entering again in Rua Dr. Alfredo de Freitas (7 houses = 1P), Rua Cruz Nova (7 houses, 2 shops, 5 apartment blocks and 4 warehouses = 2 M + 1G), Rua Chaves (2 apartment blocks, 7 houses, 1 car dealer and 1 warehouse = 2 M + 1G), = Total 14 (8P + 4 M + 2G) (Path 1–2), passing through the roundabout, entering in Rua Entre Vinhas (4 warehouses, 1 supermarket, 1 car dealer, 4 houses = 2P + 2 M + 2G), Estrada de Eiras (4 warehouses, 1 car dealer, 22 commercial buildings, 2 apartment blocks, 8 houses = Total 15 (3P + 2 M + 4G) (Path 2–5), N1, IC2 = Total 0, roundabout, N336 (1 shop =1 M) = Total 1 (1 M) (Path 8–11), roundabout, Rua Coimbra = Total 5 (3P + 2 M) (Path 11–13), entering Rua da Riqueira (17 houses = 3P + 2 M + 1G), Rua Sr. da Rua (13 houses = 2P + 1 M), Rua Santo Antônio (9 houses = 1G) = Total 10 (5P + 3 M

+ 2G)(Path 13–15), Rua do Calço (10 houses = 1 M + 1G), Rua Alta das Maias (2 houses = 1P + 1 M) (Path 15–14), arriving to ERSUC = Total 0 (Path 14–16).
Route 7: corresponds to another shortest path with 8.8 km. The route begins in União das Freguesias de Eiras and São Paulo de Frades, at Rua Dr. Alfredo de Freitas (14 houses = 1P), following through Rua 25 de Abril (22 houses =2P), Rua Viseu (22 houses = 2P), Rua Rego (10 houses = 1P), Travessa Rego (2 houses = 1P), entering again in Rua Dr. Alfredo de Freitas (7 houses = 1P), Rua Cruz Nova (7 houses, 2 shops, 5 apartment blocks and 4 warehouses = 2 M + 1G), Rua Chaves (2 apartment blocks, 7 houses, 1 car dealer and 1 warehouses = 2 M + 1G), = Total 14 (8P + 4 M + 2G) (Path 1–2), passing through the roundabout, entering in Rua Barca Ribeira de Eiras (2 supermarkets, 8 houses, 4 apartment blocks, 2 warehouses, 1 shop = 1 M + 2G) = Total 3 (Path 2–4) passing through other roundabout, entering Estrada de Eiras (14 apartment blocks, 2 shop, 10 warehouses, 2 car dealers, 1 supermarket, 4 houses = Total 11 (3P + 2 M + 6G)(Path 4–7) going through another roundabout, Rua Principal de Alcarraques (1 car dealer, 23 houses, 1 warehouse = 2 M + 3G), Estrada de Belo Monte (47 houses = 1 M + 3G) = Total 9 (3 M + 6G)(Path 7–10), Rua da Quinta da Zombaria = Total 0 (Path 10–12), roundabout at Quinta da Bolregão (Path 12–14) arriving to ERSUC = Total 0 (Path 14–16).
Route 8: corresponds to a path with 12.3 km. The route begins in União das Freguesias de Eiras and São Paulo de Frades, at Rua Dr. Alfredo de Freitas (14 houses = 1P), following through Rua 25 de Abril (22 houses = 2P), Rua Viseu (22 houses =2P), Rua Santo Cristo (24 houses = 2P), Rua Montargil (28 houses = 1P + 1 M), Rua Cruz do Vale do Seixo (25 houses, 1 apartment blocks = 2 M + 2G) = Total 13 (Path 1–3), passing through the roundabout, Rua Vale de São Miguel (7 houses and 22 apartment blocks, 2 warehouses = 2P + 1 M + 1G), Bairro São Miguel (19 apartment blocks = 2P + 2 M + 1G), roundabout = Total 9 (Path 3–6), Estrada de Eiras (5 apartment blocks, 1 shopping centre, 5 car dealers, 7 warehouses = 3P + 3 M + 1G) = Total 7 (Path 6–5), roundabout, N1, IC2 = Total 0 (Path 5–8), roundabout, N336 (1 shop =1 M) = Total 1 (1 M) (Path 8–11), roundabout, Rua Coimbra, = Total 5 (3P + 2 M) (Path 11–13), entering in Rua da Riqueira (17 houses = 3P + 2 M + 1G), Rua Sr. da Rua (13 houses = 2P + 1 M), Rua Santo António (9 houses = 1G) = Total 10 (5P + 3 M + 2G)(Path 13–15), Rua do Calço (10 houses = 1 M + 1G), Rua Alta das Maias (2 houses = 1P + 1 M) (Path 15–14), arriving to ERSUC = Total 0 (Path 14–16).

The network representation in the forward star form, (Dias Rasteiro, 2020), is in Table 12.4 and as an oriented graph with parameters associated with the arcs in Figs. 12.1a, 12.2a and 12.3a.

Table 12.4 Network problem represented in the Forward Star Form

		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
	point	1	3	5	6	7	9	10	12	13	14	15	16	17	19	20	21	21			
	suc	2	3	4	5	6	7	8	14	5	9	10	11	10	12	13	14	14	15	16	14
TIME (min)	P	16	16	3	10	8	6	3	4	9	4	3	2	2	2	6	2	2	10	2	2
	M	8	6	2	8	8	4	3	4	6	4	6	2	4	2	4	2	2	6	2	4
	G	4	5	5	14	5	14	3	3	3	5	14	3	3	2	3	2	2	5	2	3
CONTAINER'S SIZE (kg)	P	8	8	0	5	3	3	0	0	3	2	0	0	2	0	3	0	1	5	0	1
	M	4	3	1	4	4	2	0	0	3	2	3	1	2	0	2	0	0	3	0	2
	G	2	2	2	6	2	6	0	0	1	2	6	0	1	0	0	0	0	2	0	1

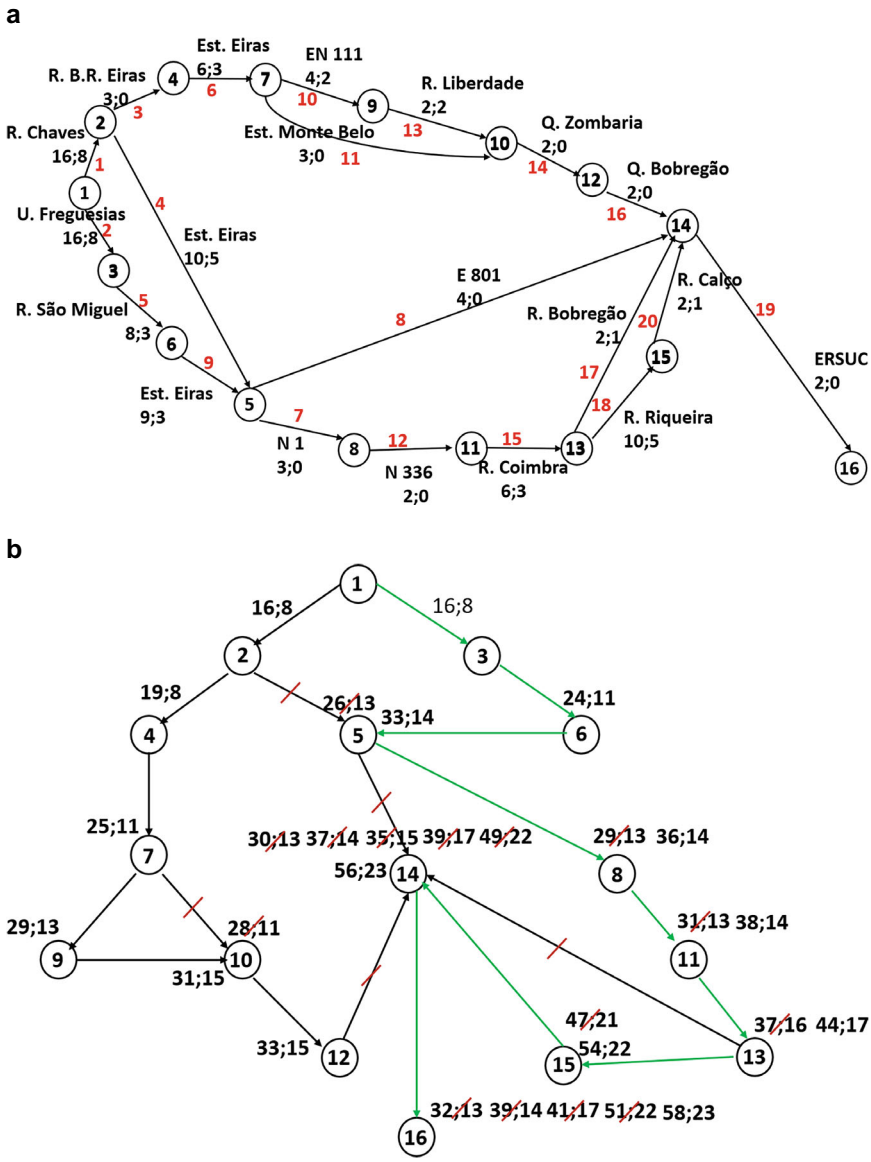

Fig. 12.1 **a** Small containers network. **b** Small containers solution; Author's own

3 Problem Solution

The network optimization problems are, in general, solved by using labelling algorithms (Martins et al., 1999). These algorithms are improvements of the exhaustive search by the optimality principles definition. To store a network, eight vectors were

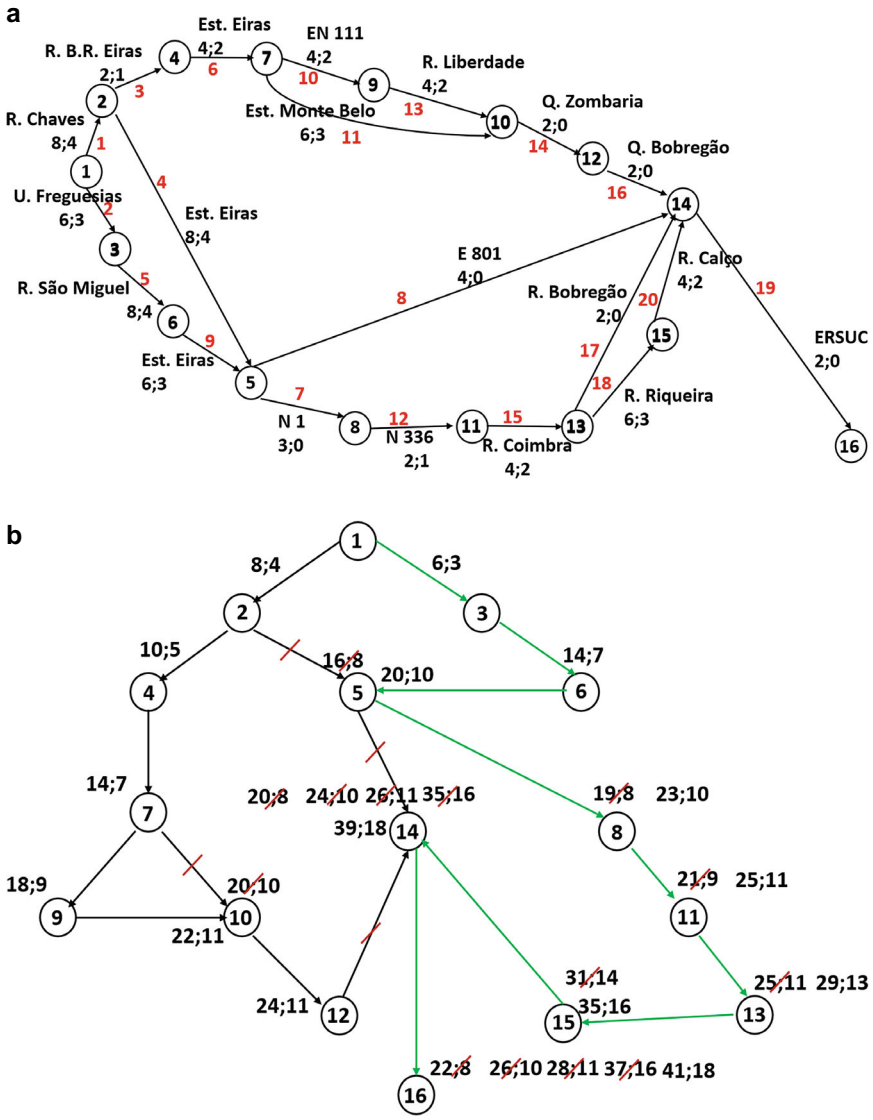

Fig. 12.2 **a** Medium containers network. **b** Medium containers solution; Author's own

considered, as can be observed on Table 12.4. The network can be stored by considering the arcs that go out from each node (forward star form) or the arcs that get in on a node (reverse forward star form) (Dial et al., 1979). We adopted the forward star form. For an exhaustive explanation of the network data storage and labelling algorithms applied, the authors invite readers to read (Dias Rasteiro, 2020). When applying

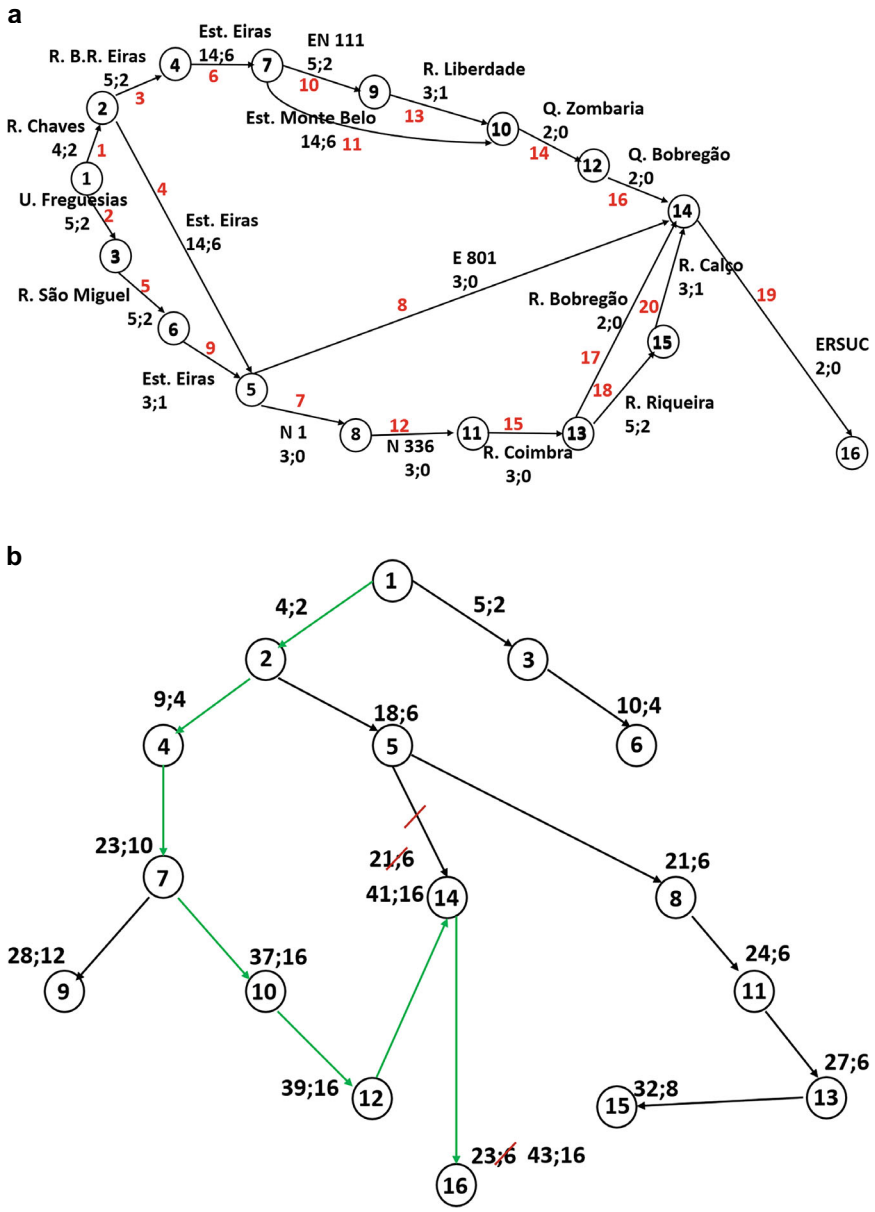

Fig. 12.3 **a** Large containers network. **b** Large containers solution; Author's own

network optimizing labelling algorithms, the following solutions, see Figs. 12.1b, 12.2b and 12.3b, are obtained.

3.1 Small Containers

L = {1,2,3,4,5,6,7,8,14,5,9,10,11,16,8,14,12,13,11,16,14,15,13,16,14,15,16,14,16}.
58 min. and 23 containers.
Optimal solution: Path 1-3-6-5-8-11-13-15-14-16.

3.2 Medium Containers

L = {1,2,3,4,5,6,7,8,14,5,9,10,11,16,8,14,12,13,11,16,14,15,13,16,14,15,16,14,16}.
41 min. and 18 containers.
Optimal solution: Path 1-3-6-5-8-11-13-15-14-16.

3.3 Large Containers

L = {1,2,3,4,5,6,7,8,14,9,10,11,16,12,13,14,15,16}.
43 min. and 16 containers.
Optimal solution: Path 1-2-4-7-10-12-14-16.

3.4 Analytic Hierarchy Process

The Analytic Hierarchy Process (AHP) is a widely utilized decision-making method designed to handle complex multi-criteria problems by structuring them into a hierarchy. This method assists in determining the weights of criteria by breaking down the decision problem into smaller, more manageable components (Saaty, 1980, 1987). Through pairwise comparisons and the use of a scale reflecting the relative importance of each criterion, AHP translates subjective assessments into numerical values. These values are then used to calculate the weights, which signify the priority of each criterion in the decision-making process. This systematic approach ensures that the decision-making process is both comprehensive and rational, aiding in the selection of the best possible alternative based on a multiple-criteria decision problem (Saaty, 2008).

Table 12.5 summarizes the aforementioned pairwise comparison for the four criteria of the decision problem under analysis and the relative importance of each criterion (weights), obtained through applying AHP.

Table 12.5 AHP Criteria pairwise comparison

	C1	C2	C3	C4	(C1*C2*C3*C4)	(C1*C2*C3*C4)^1/4	Weight %
C1	1	2	2	7	28	2,300326634	44,42
C2	0,5	1	1	5	2,5	1,25743343	24,29
C3	1	1	1	7	7	1,626576562	26,41
C4	0,142857	0,2	0,142857	1	0,004081633	0,252760077	4,88
Total						5,437096702	100

3.5 Multi-criteria Decision Analysis

After determining each criteria weight, the decision matrix, taking into account the aforementioned criteria and the three alternatives under analysis, is presented in Table 12.6.

After defining the decision matrix, a multi-criteria method is selected to solve the decision problem. The method chosen was the Simple Additive Weighting (SAW) method, as presented by Churchman and Ackoff (1954), Keeney and Raiffa (1976), and Yoon and Hwang (1995). The SAW is a straightforward and popular technique used in multi-criteria decision analysis. This method involves assigning weights to each criterion based on their relative importance and then evaluating each alternative by summing the weighted performance ratings across all criteria. The alternative with the highest total score is considered the best option. SAW is appreciated for its simplicity and ease of use, making it suitable for various decision-making scenarios where multiple criteria must be considered. It requires the decision-maker to have a clear understanding of the criteria and their respective importance, ensuring a transparent and rational evaluation process.

To obtain the results, a web calculation platform, MATRIX—A web-based multi-criteria decision support system (available at https://matrix.dec.uc.pt), was used. MATRIX is a web-based multi-criteria decision support system that offers a base of methods through a web interface for interactive editing of problems (impact matrix, parameters), centralizing multiple multi-criteria algorithms, and displaying graphical and alphanumeric results.

Table 12.6 AHP Decision Matrix

	Capacity (L)	Price total (€)	Time (min)	Maintenance total (€)
Large container	800	13,600	53	380
Medium container	500	4180	41	110
Small container	75	2400	58	100

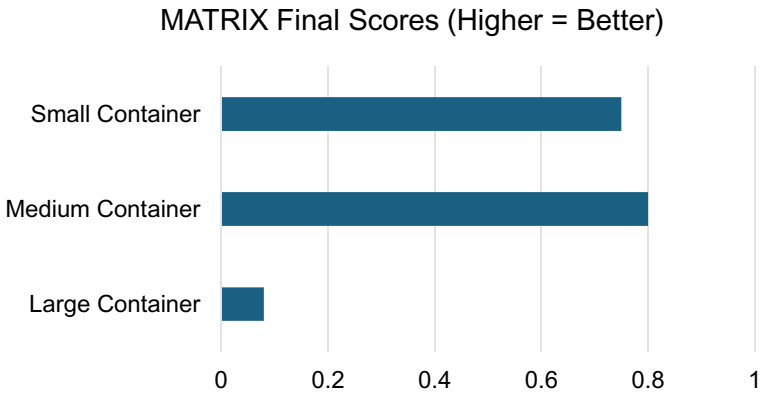

Fig. 12.4 Matrix results; Author's own

3.6 Results

Using MATRIX, and after all the data has been entered, the decision matrix, the weight vectors, and the graph with the final result (ranking) can be seen in Fig. 12.4, and it can be concluded that the medium container is the best decision for the problem at hand.

4 Clustering Analysis

For this project component, a public dataset on Municipal Waste Management[1] was used. It consists of 4341 instances and 39 attributes. Only a subset of attributes was selected to be included in the present study. These include: cost per capita in euros (tc), residual cost per capita (cres), sorted cost per capita (csor), the area in km^2 (area), population (pop), altitude mean sea level (alt), population density in people per km^2 (pden), waste per km^2 (wden), urbanization index in the scale 1 meaning low to 3 meaning high (urb), sorted municipal solid waste in kg (msw_so), unsorted municipal solid waste in kg (msw_un), municipal solid waste in kg (msw), share of sorted waste (sor), km of roads within the municipality (roads), share of solid waste sent to Waste-to-Energy plants (s_wteregio), share of waste to landfill (s_landfill), people per km of roads (proads). Many other columns contained undocumented or hard-to-obtain information for Portuguese cities, such as "d_fee" (dmmy PAYT) and "sample" (Reg with PAYT). To ensure data quality, outliers that were more than three times the standard deviation were removed. In addition, missing values were replaced with the mean of the respective column. To normalize the distributions, a Standard Scaler was applied, transforming the means to 0 and the standard deviations

[1] https://www.kaggle.com/datasets/shashwatwork/municipal-waste-management-cost-prediction.

to 1. Figure 12.5 shows two histograms of the attributes available in the dataset. On the left are the originals, on the right after they have been resized to average 0 and due to default 1.

Several clustering algorithms were tested, including K-Means, Agglomerative Clustering, Gaussian Mixture Model (GMM), and Ordering Points to Identify the Clustering Structure (OPTICS). The Elbow Plot technique with Within-Cluster Sum of Squares (WCSS) was used to determine the ideal number of clusters. K values of 5 to 11 were tested, which seemed to capture most of the information contained in the data.

Based on the silhouette score, the K-Means algorithm with k = 5 was selected as the most suitable, with a score of 0.1488. In addition, the intra- and inter-cluster distances for K-Means were evaluated, being approximately 4.12 and 5.96, respectively.

In Fig. 12.6, the result of the K-Means clustering algorithm with k = 5 projected in two dimensions, are presented. The vertical axis is the residual cost per capita (cres); horizontal axis represents the share of sorted waste (sor). Each dot is coloured with the colour of its cluster. Data seems to point to a relationship between these two variables: as the share of sorted waste increases, the residual cost per capita reduces.

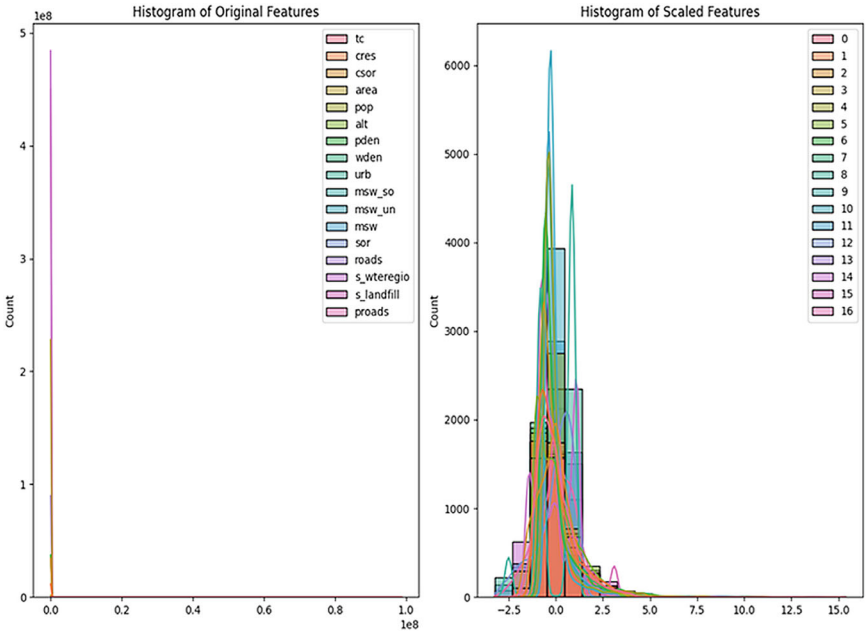

Fig. 12.5 Histograms of dataset attributes. Left: Original histograms of selected attributes from the municipal waste management dataset. Right: Histograms after applying standard scaling, transforming the means to 0 and standard deviations to 1; Author's own

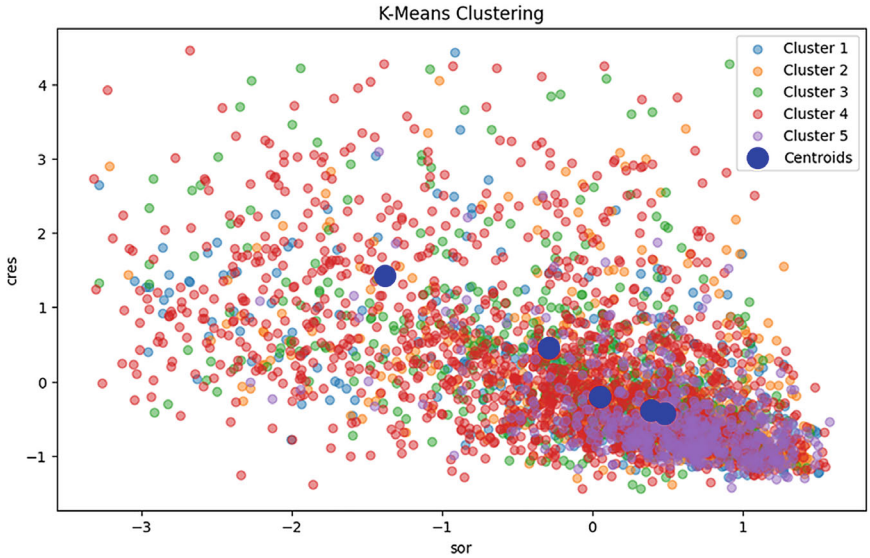

Fig. 12.6 K-means clustering result; K-Means clustering algorithm with k = 5 projected in two dimensions, where the vertical axis represents the residual cost per capita (cres), and the horizontal axis represents the share of sorted waste (sor). Each dot is coloured according to its cluster; Author's own

5 Conclusions

This study demonstrates how Problem-Based Learning (PBL) can serve as an effective educational strategy for fostering engagement and empowering students with sustainable practices aligned with SDG 12—Responsible Consumption and Production. By involving students in real-world challenges related to solid waste transportation logistics, the project has enhanced their ability to analyze, optimize, and implement sustainable solutions. The use of multi-criteria decision-making models has provided students with valuable insights into how to balance economic, environmental, and logistical factors in sustainability-oriented decision-making. The findings suggest that collaborative and interdisciplinary learning experiences play a key role in shaping students' perceptions of sustainability and responsible consumption. The active participation of master's students from the Polytechnic University of Coimbra, combined with service-learning methodologies, has fostered critical thinking, problem-solving, and decision-making skills that extend beyond the classroom. The students' engagement in waste management optimization has not only reinforced their academic knowledge but also encouraged a more profound sense of social and environmental responsibility. Despite the project's success, several challenges and limitations remain. A lack of diverse organizational structures and broader community acceptance has somewhat restricted the project's scalability. Addressing these limitations will require stronger institutional support, community outreach

initiatives, and further collaboration with policymakers and stakeholders. Looking ahead, future developments could include expanding the project's scope by integrating additional technologies such as Artificial Intelligence (AI)-based route optimization, IoT sensors for real-time waste monitoring, and blockchain for transparent waste tracking. Additionally, involving students from different disciplines, such as economics, urban planning, and environmental science, could further enhance the multidimensional approach to sustainability challenges.

In conclusion, this project highlights that education plays a fundamental role in driving sustainable practices, and by equipping students with the necessary tools, knowledge, and problem-solving skills, it is possible to foster responsible consumption and production behaviours that contribute to the broader Sustainable Development Goals (SDGs).

6 Practical Ideas

Building on the insights from this study, several practical applications and future initiatives can be implemented to enhance waste management efficiency and promote sustainability. One potential development is the creation of a smart waste management system using IoT sensors to monitor bin capacity in real-time, optimizing collection routes and reducing unnecessary trips. Additionally, mobile applications could be developed to engage citizens in responsible waste disposal, providing alerts for waste collection schedules and incentives for proper recycling habits. Another practical extension is the integration of Artificial Intelligence (AI)-driven decision-making tools to continuously refine logistics models, enhancing efficiency in solid waste transportation. The project could also inspire interdisciplinary collaborations, bringing together experts in engineering, data science, and urban planning to develop comprehensive waste management policies. Moreover, educational institutions could adopt similar PBL approaches to address other environmental challenges, such as water conservation, energy efficiency, and circular economy initiatives, reinforcing the role of students as active contributors to sustainable solutions. Lastly, local governments and waste management companies could leverage the findings of this research to design policy interventions that promote public-private partnerships, incentives for sustainable practices, and long-term community engagement strategies.

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